Pulse-type detectors
basic counting systems for, 249, 249f, 250f
pulse generation in, 247–248, 247f
pulse shape and dead time for, 250–253, 250f, 251f
Purex separation scheme, 712f, 716–719
Purity, radiochemical, 528–529
PWRs, *see* Pressurized water reactors

Q

Q value of reaction, 89, 298–299
for alpha decay, 155
closed energy cycles and, 106–107, 108f
of electron capture, 96
of internal conversion, 100
of neutron decay, 95
of positron emission, 95–96
for spontaneous fission, 159
QFT, *see* Quantum field theory
Quadrupole moment, for select nuclides, 137t
Quantum chromo dynamics, 27
Quantum field theory (QFT), antimatter and, 23
Quantum states, introduction of, 20
Quarks, 26–29
classification of, 28t
of hadrons, 27
history of, 26–27
neutron decay, 29f
in nuclear reactions, 27
properties of, 28t
Quasars, 343

R

R, *see* Measured count rate, *see* Roentgen
Ra, *see* Radium
Rad (radiation absorbed dose), 214
Radial quantum number, *see* Azimuthal quantum number
Radial velocity, 345–346
Radiation annealing, 217
Radiation background, 469–473, 468t–469t
Radiation biology
concepts of, 454f, 454t
dose–effect curve, 473f
large dose regularities, 450t, 458f–459f, 459–465
radiomimetic substances, 468–469
regulatory recommendations and protection standards, 475
cancer risk with radiation, 476–476, 475t
classifications, working rules, etc., 484f–484f, 485–489
collective dose, 473f, 477–478
committed dose, 478f, 478–482
control of, 489–490
internal radiation and dose coefficients, 480t
radiotoxicity and risks, 475t, 483f, 483–485
recommended dose limits, 476
Radiation chemical yield, 216
Radiation chemistry, 164, 232–235
in biological systems, 450t, 452–453
industrial radiation processing, 234–235
process criteria, 233–234
radiation sources for, 232–233, 233f
radiation-induced synthesis, 234
Radiation chemists, 446
Radiation dose, 214–216, 215f
biological effect of, 446
Radiation effects, 574
Radiation effects on matter, 209–238
aqueous solutions, 223–225, 224f
dose measurements, 229–232, 230f
energy transfer, 210–212
charged particles, 210–212, 211t
uncharged radiation, 212
experimental methods, 228–229
inorganic nonmetallic compounds, 217–219, 218f–219f
metals, 216–217
nonbiological applications of, 232–235
industrial radiation processing, 234–235
process criteria, 233–234
radiation sources for, 232–233, 233f
radiation-induced synthesis, 234
organic compounds, 225–228, 225t, 227t
radiation dose and yield, 214–216, 215f
radiation tracks, 212–214, 213f
small dose rates, 235–236
water, 219–222, 220f, 221t, 222f
Radiation physics, 450–452
Radiation protection agents, 459
Radiation resistance, 459